# UID Exp: 10

## AIM:

The aim of this experiment is to create data visualizations such as pie charts, bar charts, and line charts for an **Attendance Monitoring System** using JavaScript.

## PROCEDURE:

### Step 1: Set Up Your HTML File

Create an HTML file (index.html) to design the structure of the Attendance Monitoring System. Include:

1. A title **"Attendance Monitoring System"**
2. Canvas elements for Pie Chart, Bar Chart, and Line Chart
3. Link the Chart.js library
4. Link the external JavaScript file (script.js)

index.html × script.js

index.html > html > body > p > br

```html
<!DOCTYPE html>
<html>
<head>
<title>Attendance</title>
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<style>
body{text-align:center;font-family:Arial}
.box{display:inline-block;width:200px}
canvas{width:180px!important;height:180px!important}
</style>
</head>
<body>

<h2><b>Attendance Monitoring System</b></h2>

<div class="box"><p><b>Attendance Status Distribution</b></p><canvas id="p"></canvas></div>
<div class="box"><p><b>Daily Attendance Volume</b></p><canvas id="b"></canvas></div>
<div class="box"><p><b>Attendance Trends over Time</b></p><canvas id="l"></canvas></div>

<p><b>Features</b><br>
✓ Present vs Absent<br>
✓ Daily Data<br>
✓ Trend Analysis<br>
✓ Easy Visualization</p>

<script src="script.js"></script>
</body>
</html>
```

## Step 2: Create the JavaScript File

Create a JavaScript file (script.js) to implement the chart logic.

1. Define attendance data (days and percentage)
2. Create a **Pie Chart** to show present vs absent
3. Create a **Bar Chart** to display daily attendance
4. Create a **Line Chart** to show attendance trends over time

index.html  script.js

script.js > lineChart > data

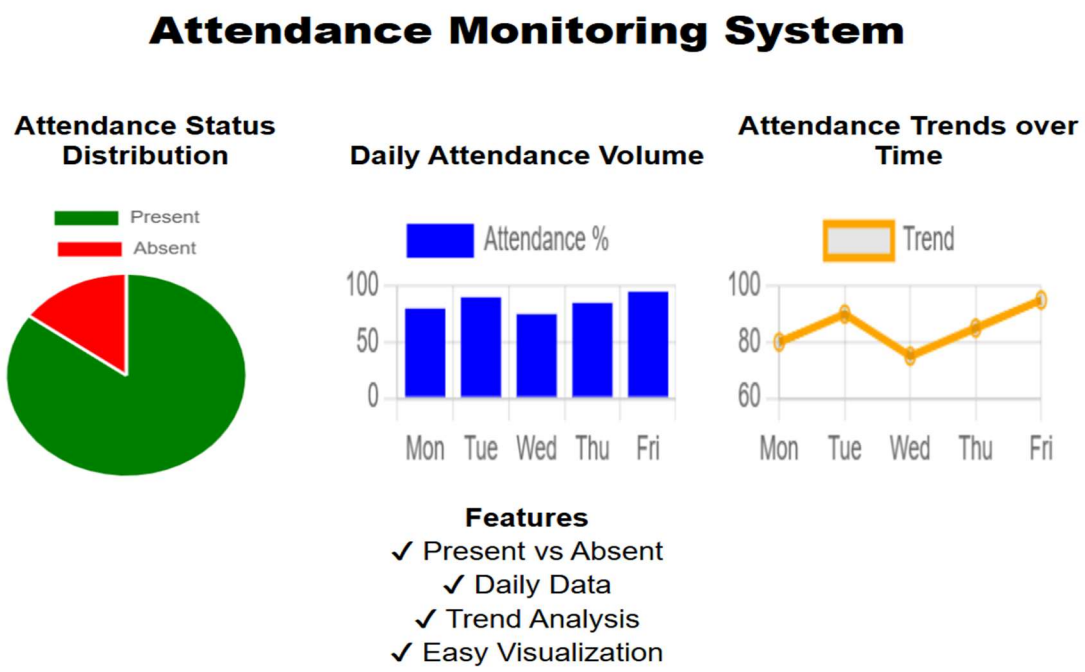
```javascript
const days = ["Mon", "Tue", "Wed", "Thu", "Fri"];
const attendance = [80, 90, 75, 85, 95];
const pieChart = new Chart(document.getElementById("p"), {
    type: 'pie',
    data: {
        labels: ["Present", "Absent"],
        datasets: [{
            data: [85, 15],
            backgroundColor: ["green", "red"]
        }]
    }
});
const barChart = new Chart(document.getElementById("b"), {
    type: 'bar',
    data: {
        labels: days,
        datasets: [{
            label: "Attendance %",
            data: attendance,
            backgroundColor: "blue"
        }]
    }
});
const lineChart = new Chart(document.getElementById("l"), {
    type: 'line',
    data: {
        labels: days,
        datasets: [{
            label: "Trend",
            data: attendance,
            borderColor: "orange",
            fill: false
        }]
```

## Features:

1. Displays present and absent percentage
2. Shows daily attendance data
3. Tracks attendance trends over time
4. Simple and user-friendly visualization
5. Helps in monitoring student performance

## Output:

The system displays:




1. A **Pie Chart** representing attendance status distribution
2. A **Bar Chart** showing daily attendance volume
3. A **Line Chart** representing attendance trends over time
4. All charts are displayed clearly on a single web page

## Result:

The Attendance Monitoring System was successfully implemented and the charts were displayed correctly, helping to visualize attendance data effectively.